

Essential SSD Specifications to Verify

A. Sequential Read/Write

Sequential Read/Write measures how quickly an SSD can read or write large blocks of data stored in consecutive logical addresses. Think of an SSD as a long row of storage locations numbered 1, 2, 3, 4, 5, and so on. During a sequential operation, the SSD accesses these locations in continuous order. For example, during a sequential read, data may be accessed as $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7$, while a sequential write stores data in a similar continuous sequence. This is different from **Random Read/Write**, where the SSD accesses data from scattered locations, such as $3 \rightarrow 829 \rightarrow 41 \rightarrow 602 \rightarrow 17 \rightarrow 950$. Sequential operations are particularly important when working with large files, such as videos, backups, disk images, and large file transfers. Sequential performance is normally measured in **MB/s (megabytes per second)** or **GB/s (gigabytes per second)**.

Sample software – [CrystalDiskMark](#)

CrystalDiskMark creates or uses a **benchmark test file** on the selected drive and performs controlled read/write operations against it. It measures the amount of data transferred and the time required, then reports throughput.

B. Random Read/Write IOPS

- **Random Read/Write IOPS** tells you how quickly an SSD can handle **many small pieces of data located at different storage addresses**.
- Unlike sequential testing, which accesses data continuously:
 - Sequential:
 $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6$
 - Random testing accesses scattered locations:
 - Random:
 $4 \rightarrow 927 \rightarrow 31 \rightarrow 650 \rightarrow 102 \rightarrow 845$
- This is very important because operating systems, applications, databases and games frequently access many small files or small pieces of data rather than one huge continuous file.

C. TBW (Endurance)

TBW = Terabytes Written.

TBW is an **endurance rating** that tells you how much data the SSD is warranted/rated to have written to it over its specified endurance period. It is mainly related to the finite program/erase endurance of NAND flash.

For example, if a 1 TB SSD is rated at:

600 TBW

it means the endurance specification allows for approximately **600 TB of host writes** under the manufacturer's stated conditions. It does **not** mean the SSD will automatically stop working as soon as it reaches 600 TBW.

D. Sustained Write Speed

Sustained Write Speed tells you how fast an SSD can **continue writing a large amount of data for an extended period**, especially after its high-speed SLC cache becomes full.

SLC Cache is a high-speed temporary storage area in an SSD that allows incoming data to be written faster before it is moved to the main TLC or QLC NAND storage.

This is different from the peak write speed you normally see on an SSD box.

For example:

Advertised Sequential Write: Up to 3,000 MB/s

Sustained Write after cache: 700 MB/s

Both numbers can be valid—they measure different behavior.

E. Temperature

Purpose:

- Detect thermal issues and throttling.

Ideal:

- Idle: 30–50°C
- Load: Below 75°C

Warning:

- Above 80°C may cause throttling.